

Yanjia Huang

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Summary

Research Interests: My research focuses on **Embodied AI** and **World Models** with two thrusts: (1) **Manipulation** learning skills from video with explicit spatial reasoning for long-horizon tasks; (2) **Navigation** building predictive world models that imagine and evaluate future states to select the next step like human.

Highlight: 4 years of research experience with solid mathematical and theoretical background; 3 years of experience with real robotic platforms: Kinova gen2, Franka and Realman compound robot.

Education

Texas A&M University

M.S. IN ENGINEERING TECHNOLOGY

College Station, Texas, United States

Aug. 2024 - Present

Liverpool John Moores University

B.ENG.(HONS) IN MECHANICAL ENGINEERING (FIRST CLASS HONOURS)

Liverpool, England

Sep. 2020 - Jun. 2024

University of Shanghai for Science and Technology

B.ENG. IN MECHANICAL ENGINEERING

Shanghai, China

Sep. 2020 - Jun. 2024

Publications

(*equal contribution)

[1] Zero-shot Object Navigation with Vision-Language Models Reasoning

ICPR 2024

CONGCONG WEN^{1*}, YISIYUAN HUANG^{2*}, **YANJIA HUANG^{2*}**, WENYU HAN², SHUAIHANG YUAN¹, YU-SHEN LIU³ AND YI FANG¹

[2] A self-healing composite actuator for multifunctional soft robot via photo-welding

Composites Part B: Engineering

MINGXIA LIU^{1,2}, SHU ZHU³, **YANJIA HUANG¹**, ZIHUI LIN², WEIPING LIU^{1,4}, LILI YANG¹ AND DENGTEG GE²

[3] Can Large Vision Language Models Read Maps like a Human?

NeurIPS 2025 (under review)

XING SHUO^{1*}, SHUANGYU XIE^{2*}, ZEZHOUSUN^{3*}, KAIYUAN CHEN², **YANJIA HUANG¹**, YUPING WANG⁴, JIACHEN LI⁵, DEZHEN SONG³, ZHENGZHONG TU¹

[4] PANDORA: Diffusion Policy Learning for Dexterous Robotic Piano Playing with Oracle Reward Assessment

NeurIPS 2025 Workshop, IROS 2025

YANJIA HUANG¹, RENJIE LI¹, ZHENGZHONG TU¹

[5] VISTA: Visual Imagination with Scheduler for Task-Aware Navigation

CoRL 2025 (under review)

YANJIA HUANG¹, MINGYANG WU¹, RENJIE LI¹, ZHENGZHONG TU¹

[6] Lock, Forecast, Act: End-to-End Object-Centric 3-D Gaussian Splatting and Diffusion Trajectory Prediction for Dynamic Grasping

ICCV Workshop 2025

YANJIA HUANG^{1*}, MINGYANG WU^{1*}, RUIJIE YE², RENJIE LI¹, ZHENGZHONG TU¹

[7] FORGE-Tree: Diffusion-Forcing Tree Search for Long-Horizon Robot Manipulation

ICRA 2026 (under review)

YANJIA HUANG^{1*}, SHUO LIU^{2*}, SHENG LIU³, QINGXIAO XU¹, MINGYANG WU¹, XIANGBO GAO¹, ZHENGZHONG TU¹

[8] VISTAv2: World Imagination for Indoor Vision-Language-Navigation

ICRA 2026 (under review)

YANJIA HUANG^{1*}, XIANGBO GAO¹, XIANSHUN JIANG³, MINGYANG WU¹, ZHENGZHONG TU¹

[9] MXene-Based Hydrogel Biosensors for Oral Inflammation and Lump Detection

NSF Proposal

WEIJIA LIU¹, **YANJIA HUANG¹**, ZHENGZHONG TU¹, CHENGLIN WU¹

Research Experience

UCLA

RESEARCH ASSISTANT, AIVC LAB

Los Angeles, CA, U.S.A

Nov. 2025 - Present

- **Research Topics:** Deformable Object Manipulation
- Building a robotic origami system that learns folding from human videos, enabling few-shot generalization to new shapes.

Reality.Inc

ML AGENT TEAM TECH LEAD, SCAM.AI

Berkeley, CA, U.S.A

April. 2025 - Present

- Successfully secured a **\$300K angel round**
- As ML Agent Team Lead, spearheading the development of AI-driven SMS phishing attack agents to validate and strengthen our defense capabilities.

Texas A&M University

RESEARCH ASSISTANT, ADVISED BY PROF. ZHENGZHONG TU

College Station, TX, U.S.A

Aug. 2024 - Present

- **Research Topics:** Robotics, Embodied Agent, Computer Vision, Vision-Language-Model(VLM)
- Developed world model for robot manipulation and navigation learning with explicit geometric reasoning, and imaginative next step planning via predictive models.

Noah's Ark Lab

RESEARCH INTERN

Shanghai, China

Oct. 2023 - Aug. 2024

- **Research Topics:** Real-World Embodied AI, Voice Command-Driven Robotics
- Creating a compound robot that can perform complex tasks like pouring water in challenging settings such as offices and can navigate to objects all through voice commands.

New York University

RESEARCH ASSISTANT AT MULTIMEDIA AND VISUAL COMPUTING LAB, ADVISED BY PROF. YI FANG

New York, U.S.A

May. 2023 - Sept. 2023

- **Research Topics:** Autonomous Vehicle Navigation, Vision-Based Navigation, Motion and Path Planning
- Proposed a novel Vision Language model named VLTNet (Vision Language model with Tree-of-thoughts NETwork) for Language-driven Zero-Shot Object Navigation (L-ZSON) and it leverages a ToT reasoning framework, enabling multi-path reasoning, self-evaluation, and anticipation of future demands.

Fudan University

RESEARCH ASSISTANT, ADVISED BY PROF. YANWEI FU

Shanghai, China

Sept. 2022 - Present

- **Research Topics:** Dexterous Robotic Hand Manipulation, Machine Learning in Robotic Control
- Contributed into a pioneering method for approximating the 3D shape of liquid using the 6-DoF pose of source containers and the estimated liquid mask. This approach is innovative in that it models 3D liquid from a single image without relying on temporal information.

Skills

Programming	C, C++, Python, SQL, Linux, MATLAB, ROS
Platforms	Kinova, Franka, Realman
Design	Solidworks, CAD, ArtCAM, COMSOL, Arduino

Honors & Awards

2021	First Prize , RoboMaster University Technical Challenge (RMUT) 2021 Regional Competition (Central Region)	Jiangsu, China
2021	First Prize (Top 5%) , Internet+ Innovation and Entrepreneurship Competition	Shanghai, China
2023	the Dean's Scholarship , Sino-British College, University of Shanghai for Science and Technology	Shanghai, China